VISUALISATIONS AND SKEW

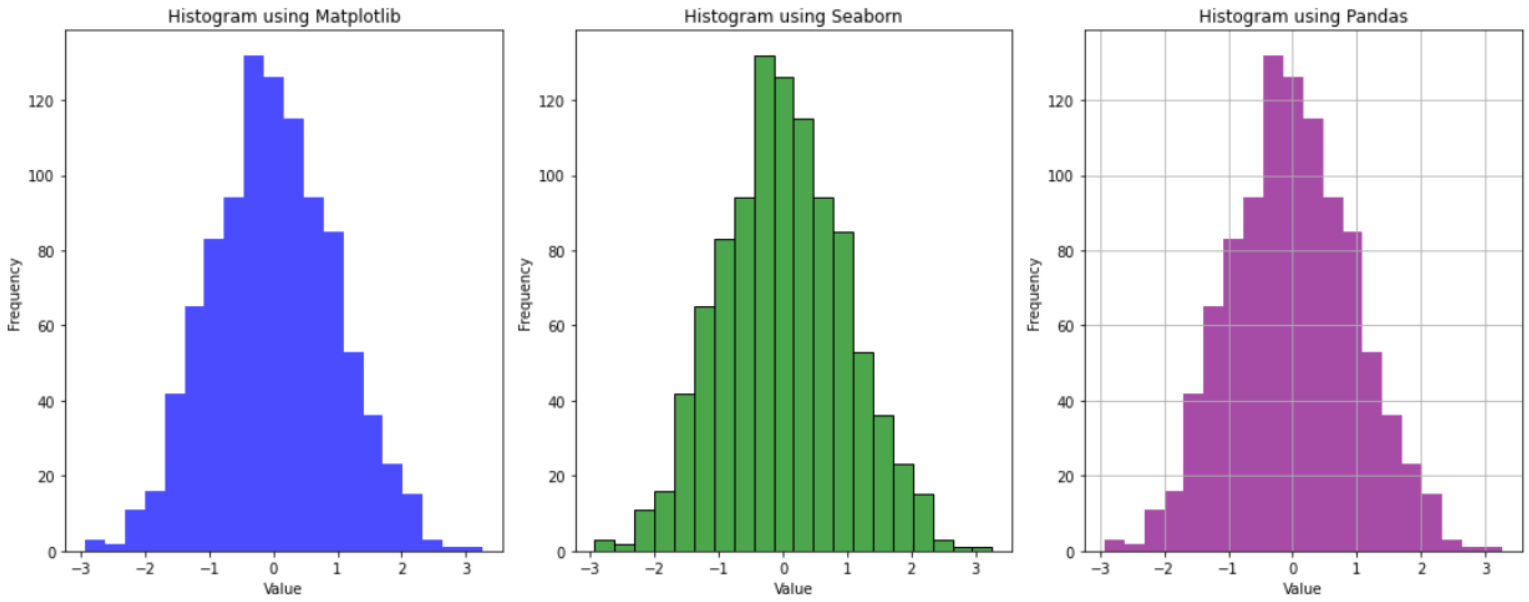
Learning objectives

- Using an appropriate software tool, create:
  - Histograms
  - Scatter Plots
- Use these to:
  - Identify skew and the effect this may have on modelling
  - Identify the location of the averages
  - Compare two samples (e.g., taken at different times or from different locations)
  - Determine the appropriate shape of a model and whether there are opportunities to linearise

Expected prior knowledge

- Experience with Python or R for Data Handling

HISTOGRAMS: PYTHON



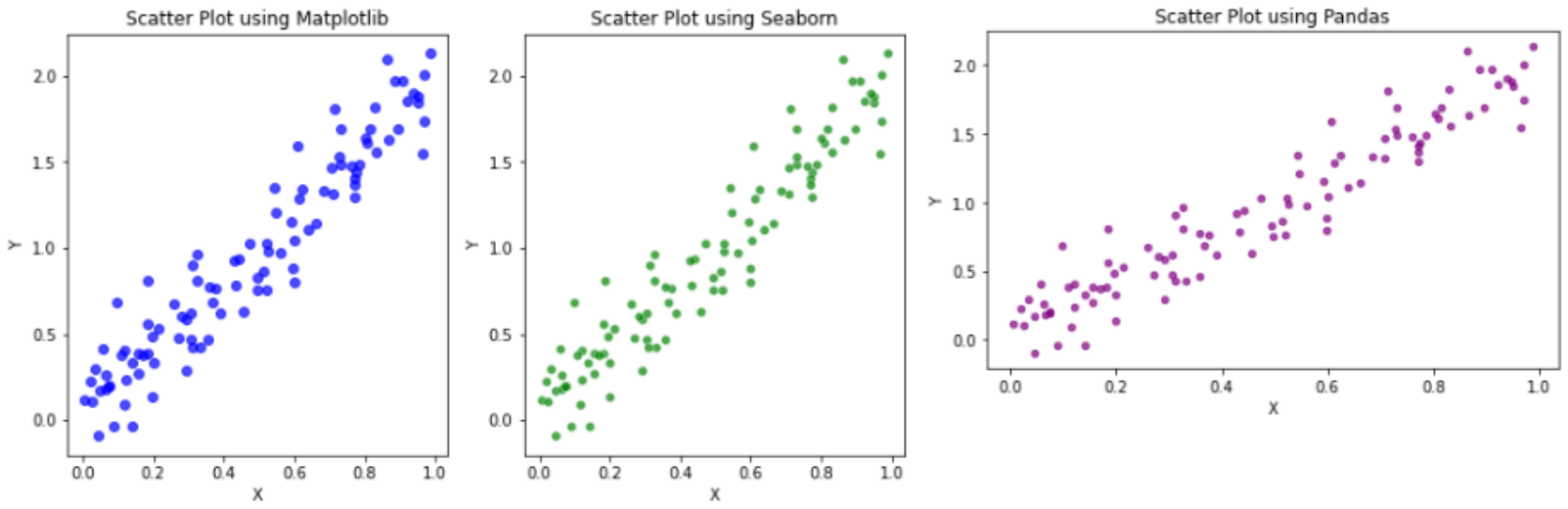

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Create a sample dataset
data = np.random.randn(1000)  # Generating random data

# Creating a Pandas DataFrame
df = pd.DataFrame({'data': data})
```

```python
plt.hist(df['data'], bins=20, color='blue', alpha=0.7)
```

```python
sns.histplot(data=df, x='data', bins=20, color='green', alpha=0.7)
```

```python
df['data'].hist(bins=20, color='purple', alpha=0.7)
```

SCATTER PLOTS: PYTHON



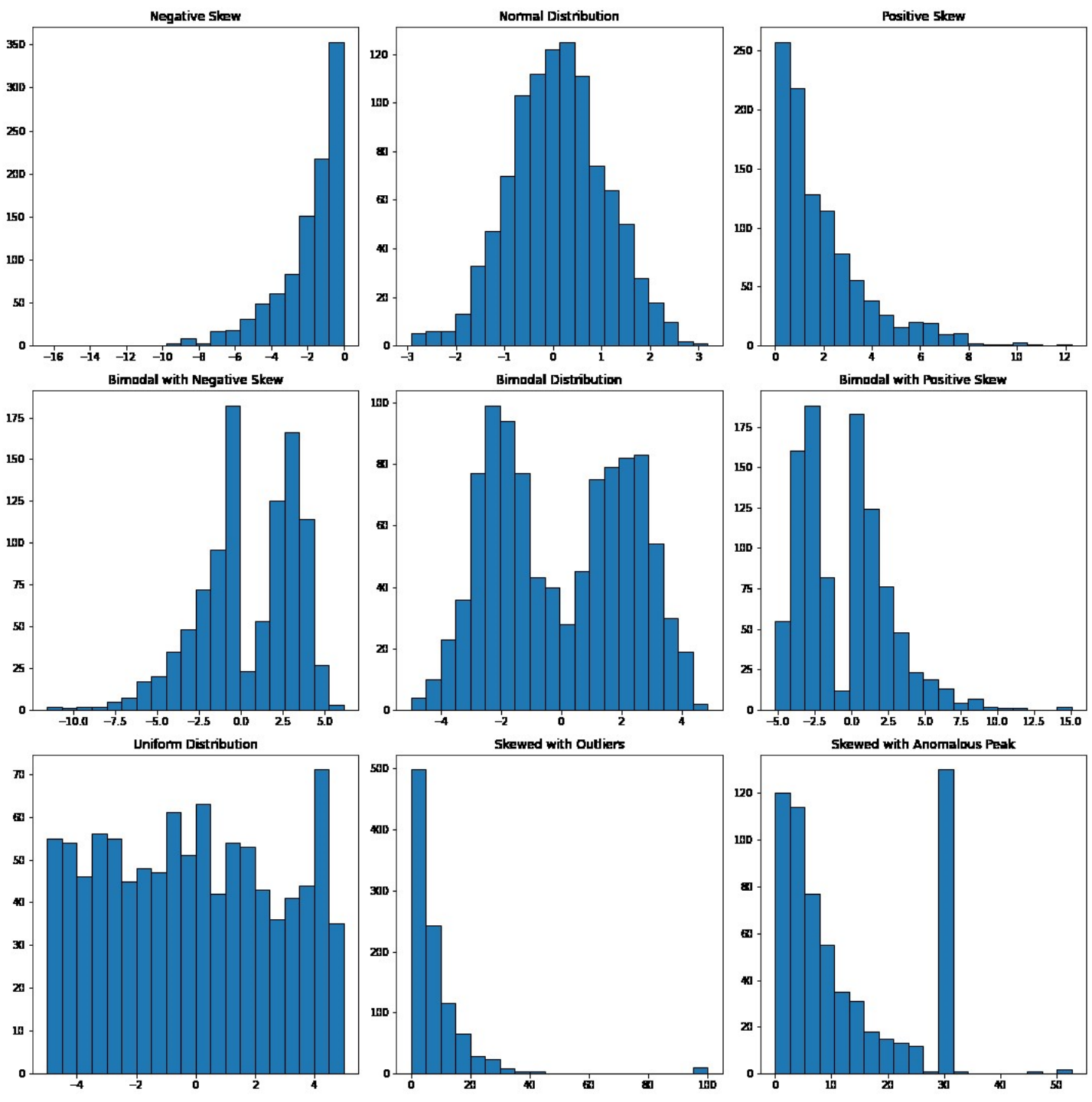

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Create a sample dataset
np.random.seed(42)
n_points = 100
x = np.random.rand(n_points)
y = 2 * x + np.random.normal(0, 0.2, n_points)

# Creating a Pandas DataFrame
df = pd.DataFrame({'X': x, 'Y': y})
```

```python
plt.scatter(df['X'], df['Y'], color='blue', alpha=0.7)
```

```python
sns.scatterplot(data=df, x='X', y='Y', color='green', alpha=0.7)
```

```python
df.plot.scatter(x='X', y='Y', color='purple', alpha=0.7)
```

DISTRIBUTION AND SKEW: VISUALLY



DISTRIBUTION AND SKEW: VISUALLY

Symmetry vs. Asymmetry:
A perfectly symmetrical distribution, like the normal distribution, has a mean, median, and mode that are all equal.

Skewness measures how much a distribution deviates from this symmetry.

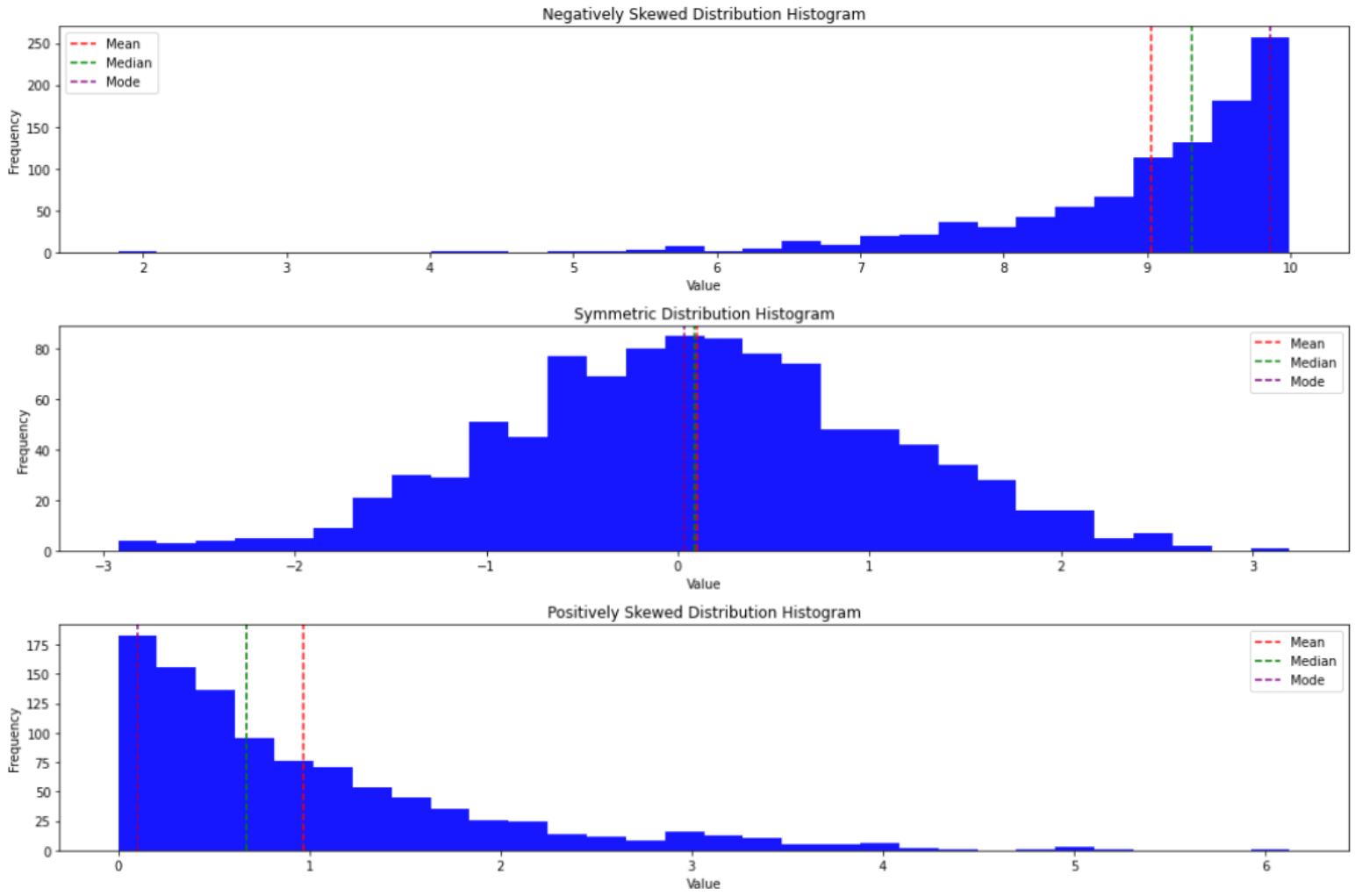
Positive Skew (Right Skew):

In a positively skewed distribution, the tail on the right side is longer or more pronounced than the tail on the left. The mean is greater than the median, and the mode is less than both.

Negative Skew (Left Skew):

In a negatively skewed distribution, the tail on the left side is longer. The mean is less than the median, and the mode is greater than both.

LOCATION OF AVERAGES IF THERE IS SKEW WHICH IS EFFECTED THE MOST?



LOCATION OF AVERAGES IF THERE IS SKEW WHICH IS EFFECTED THE MOST?

Mean and Median:
The mean (average) is sensitive to outliers and extreme values, while the median is more resistant to them. In a right-skewed distribution, the presence of a long tail of larger values pulls the mean higher than the median.

Example:
Imagine a dataset of salaries. A few very high-earning individuals (outliers) would cause the mean salary to be higher than the median salary, even if most people earn within a relatively narrow range.

Understanding the shape of a distribution can help in choosing the appropriate measure of central tendency and interpreting the data accurately. if a dataset is right-skewed, the median might be a better indicator of the "typical" value than the mean.

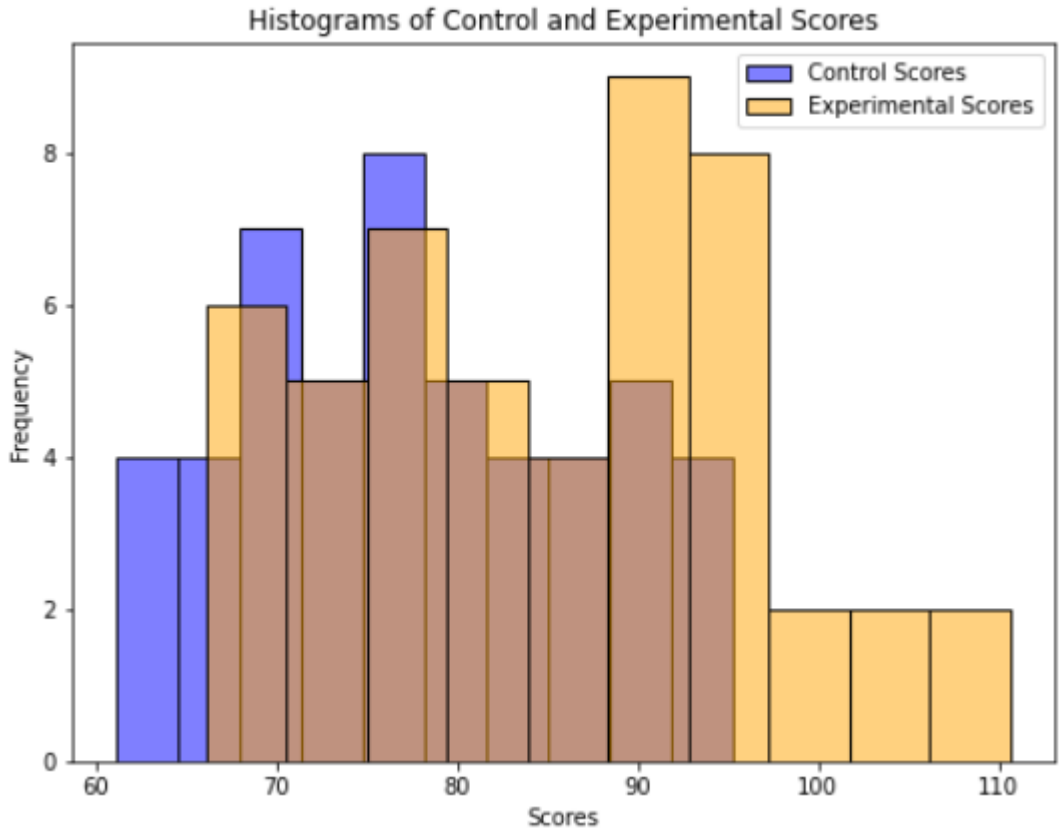
Exercise: Compare two groups using histograms

1. Import pandas as pd and use pd.read_csv() to save the investment_performance csv file as a pandas dataframe named df.
2. View the data. The scores are results from two investment strategies – the control group (using the current method) and the experimental group (using a new strategy).
3. Import seaborn as sns and matplotlib.pyplot as plt. Use part, or all of the code below to create histograms for the control and experimental groups. What does the visualisation tell you?

```
plt.figure(figsize=(8, 6))
sns.histplot(data=df, x='Control Scores', bins=10, color='blue', label='Control Scores', alpha=0.5)
sns.histplot(data=df, x='Experimental Scores', bins=10, color='orange', label='Experimental Scores', alpha=0.5)
plt.title('Histograms of Control and Experimental Scores')
plt.xlabel('Scores')
plt.ylabel('Frequency')
plt.legend()
plt.show()
```

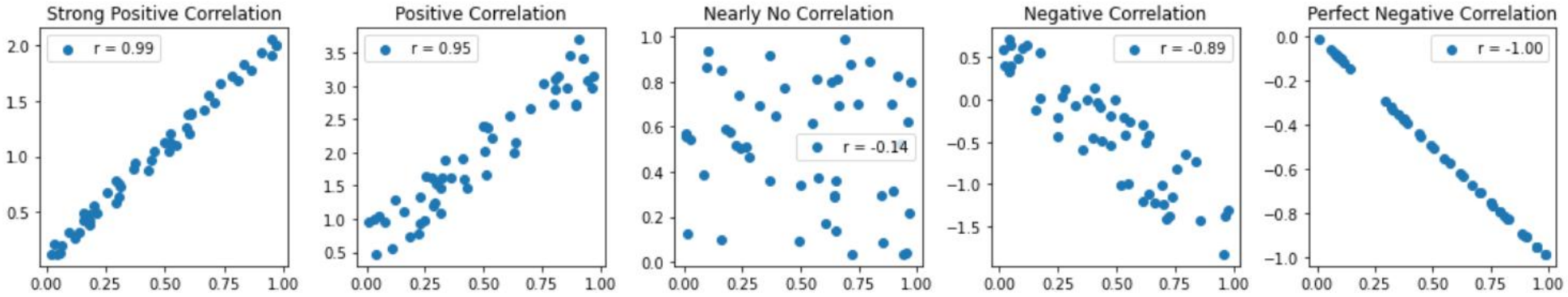
SOLUTION TO DISCUSS

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load data from the CSV file using pandas
csv_filename = 'investment_performance.csv'
df = pd.read_csv(csv_filename)
```



Correlation from scatter plots

Checking what a scatter plot looks like is another way of checking for outliers or anomalies, but we also might be looking to see if a line of best fit (or regression line) might be appropriate.

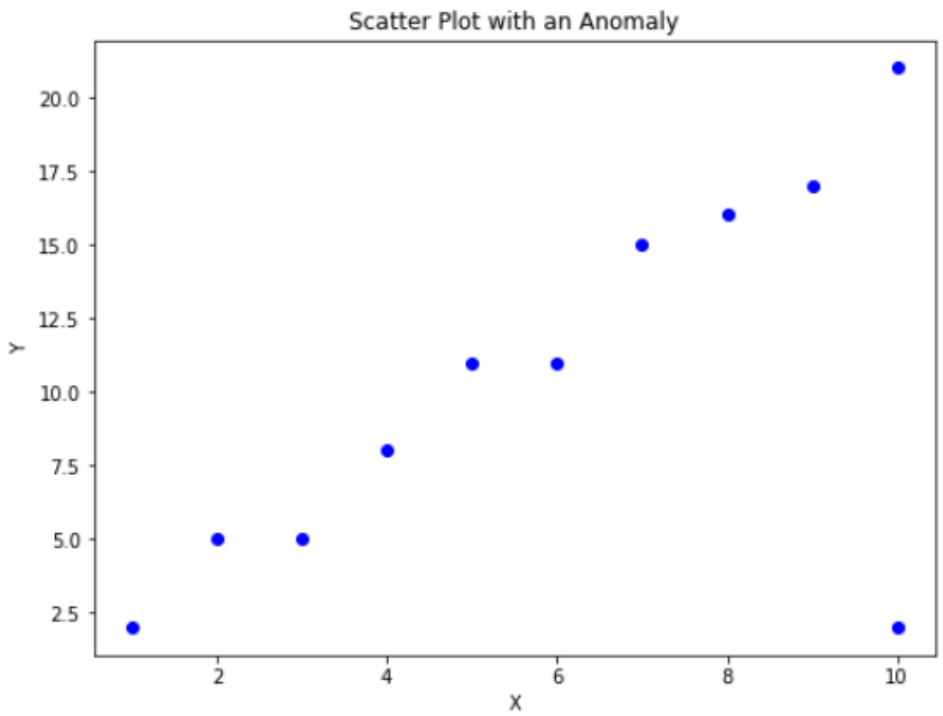


The more squashed the ‘ball’ of data is the stronger the correlation.

Correlation or association can tell us if two data features are linked, but one hasn’t necessarily caused the other – there may be a third unseen feature that is a **causal link**.

DISCUSSION: ANOMALY SPOTTING ON SCATTER PLOTS

Is there any data that looks out of place?



If the x and y coordinates for the point aren't detected as outliers... how might an algorithm work that could detect anomalies like this one?

LEARNING CHECK

Think about your answers to these questions:

- How can you create histograms and scatter plots?
- What effect does skew have on the mean, mode, or median?
- Why might it be useful to look at histograms and scatter plots?

How did you get on?

Learning Objectives

- Using an appropriate software tool, create:
  - Histograms
  - Scatter Plots
- Use these to:
  - Identify skew and the effect this may have on modelling
  - Identify the location of the averages
  - Compare two samples (e.g., taken at different times or from different locations)
  - Determine the appropriate shape of a model and whether there are opportunities to linearise